Lecture 2 (Week 1)
Chapter 1

What is Interaction Design II: Design Principles and Accessibility

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Accessibility and Inclusiveness

- **Accessibility:**
  - The extent to which an interactive product is accessible by as many people as possible
  - Focus is on people with disabilities; for instance, those using android OS or apple voiceover

- **Inclusiveness:**
  - Making products and services that accommodate the widest possible number of people
  - For example, smartphones designed for all and made available to everyone regardless of their disability, education, age, or income

Disabilities

- Whether someone is disabled changes over time with age, or recovery from an accident
- The severity and impact of an impairment can vary over the course of a day or in different environmental conditions
- Technologies are designed to necessitate a certain type of interaction that is often impossible for someone with an impairment

Understanding Disability

- Disabilities can be classified as:
  - Sensory impairment (such as loss of vision or hearing)
  - Physical impairment (having loss of functions to one or more parts of the body after a stroke or spinal cord injury)
  - Cognitive (including learning impairment or loss of memory/cognitive function due to old age)

- Each type can be further defined in terms of capability:
  - For example, someone might have only peripheral vision, be color blind, or have no light perception

Understanding Disability: Impairment

- Impairment can be categorized:
  - Permanent
    - (for instance, long-term wheelchair user)
  - Temporary
    - (that is, after an accident or illness)
  - Situational
    - (for example, a noisy environment means that a person can't hear)

Being Cool about Disability

- Prosthetics can be designed to move beyond being functional (and often ugly) to being desirable and fashionable

- People now refer to

  "wearing their wheels," rather than "using a wheelchair"

Fashionable leg cover designed by Alleles Design Studio

Design Principles

- Generalizable abstractions for thinking about different aspects of design
- The do's and don'ts of interaction design
- What to provide and what not to provide at the interface
- Derived from a mix of theory-based knowledge, experience, and common-sense

Visibility

- Can see the state of a device and possible actions
- Car controls are positioned in a way that they can be easily found and used

Visibility - Poor Interface

- This is a control panel for an elevator
- How does it work?
- Push a button for the floor you want?
- Nothing happens. Push any other button? Still nothing.
- What do you need to do?
- It is not visible as to what to do!

www.baddesigns.com

Visibility - Improving on a Poor Interface

- With this elevator, you need to insert your room card in the slot by the buttons to get the elevator to work!

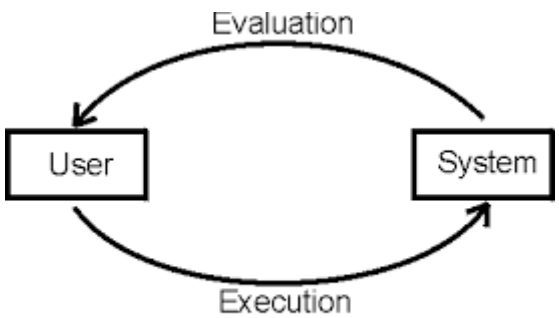
- How would you make this action more visible?
  - Make the card reader more obvious
  - Provide an auditory message that says what to do (which language?)
  - Provide a big label next to the card reader that flashes when someone enters
  - Make what has to be done obvious

Further on Visibility

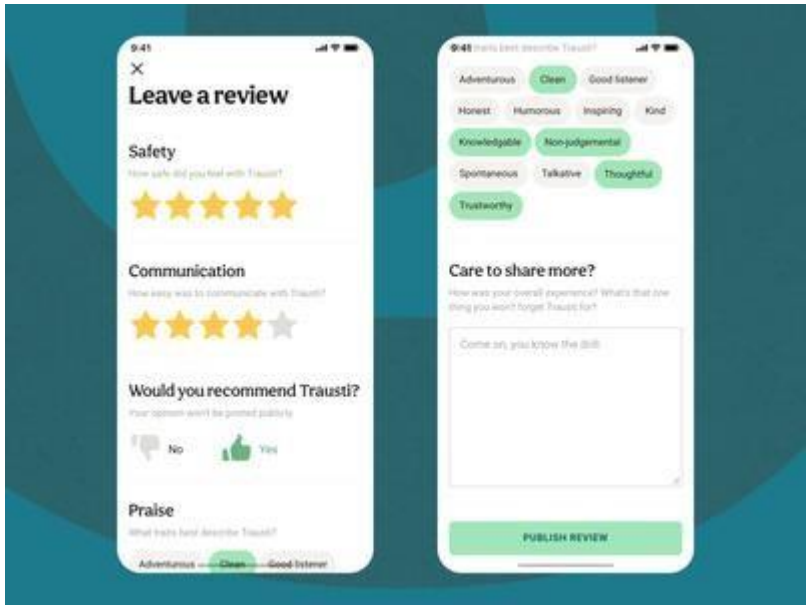
- Invisible automatic controls can make it more difficult to use

Feedback

- Sending information back to the user about what has been done
- Includes sound, highlighting, animation, and combinations of these
  - For example, when screen button is clicked, it provides sound or red highlight feedback:



Norman's Interaction Cycle



Constraints

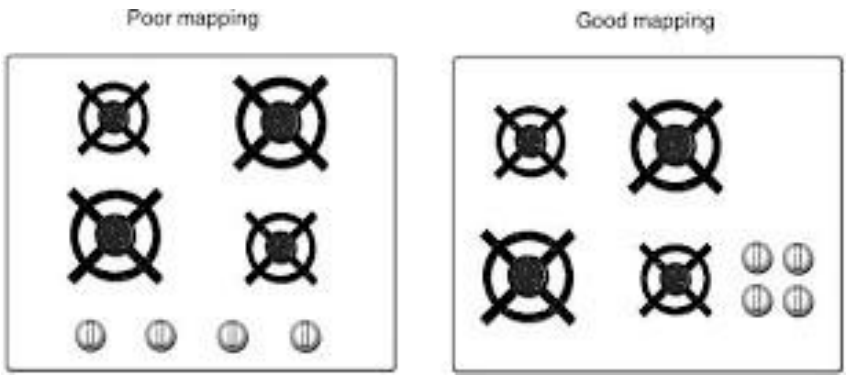
- Restricting the possible actions that can be performed
- Helps **prevent** user from selecting incorrect options
- Physical objects can be designed to constrain things. (For example, there being only one way you can insert a key into a lock)

Logical or Ambiguous Design?

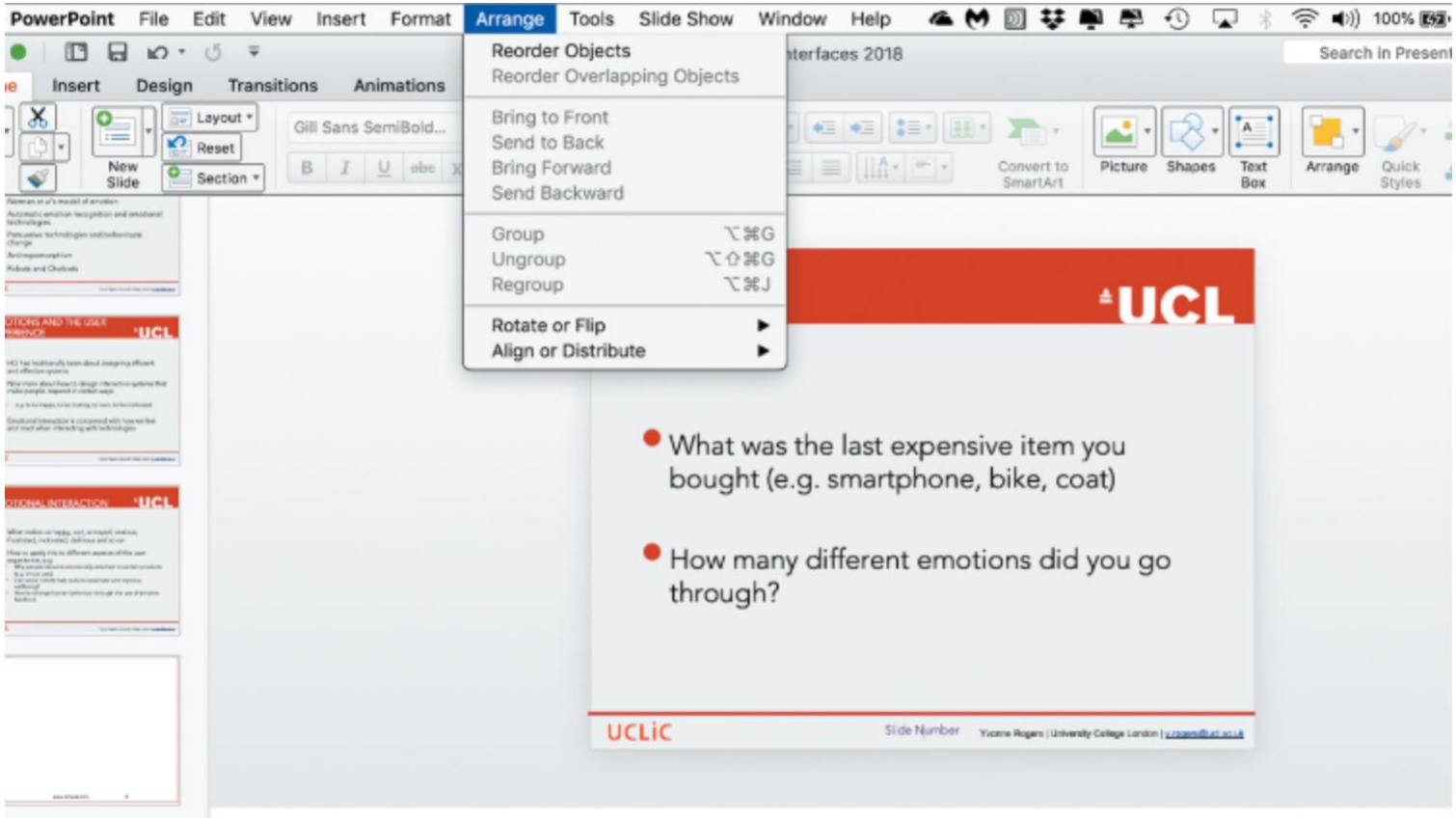
- Where do you plug the mouse?
- Where do you plug the keyboard, in the top or bottom connector?
- Do the color-coded icons help?

Mapping: How to Design Them More Logically?

- Provides direct adjacent mapping between icon and connector
- Provides color coding that associates the connectors with the labels
- Any other daily life examples of logical constraint?



Logical Constraint Example



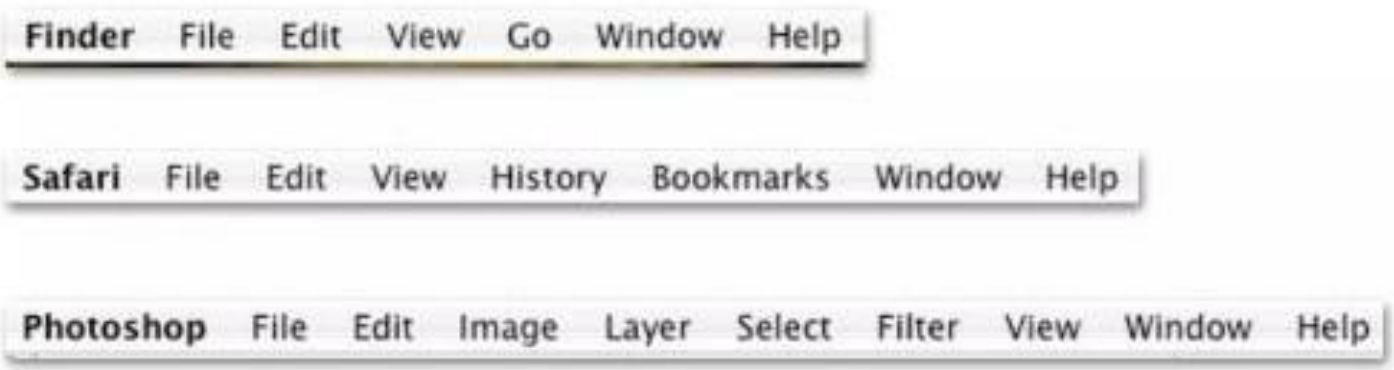
Consistency

- Design interfaces to have similar operations and use similar elements for similar tasks.
- (for example, always use Ctrl key plus first initial of the command for an operation: Ctrl+c, Ctrl+s, Ctrl+o)

- The main benefit is that consistent interfaces are easier to learn and use

- Types of consistency: aesthetic, internal, and external

Consistency



When consistency breaks down

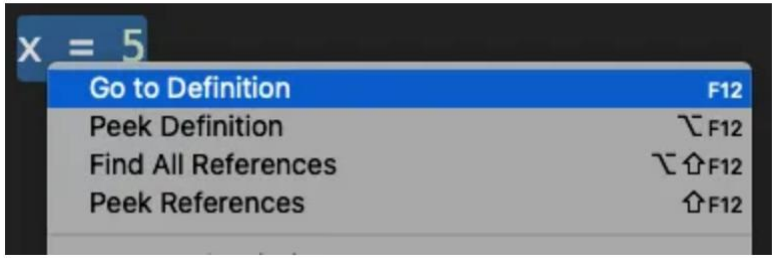
- What happens if there is more than one command starting with the same letter? (for example, save, spelling, select, style)
- You have to find other initials or combinations of keys, thereby breaking the consistency rule (for example, Ctrl+s, Ctrl+Sp, Ctrl+shift+l)
- Increases learning burden on user, making them more prone to errors

Internal and External Consistency

- Internal consistency refers to designing operations to behave the same within an application
  - Difficult to achieve with complex interfaces

- External consistency refers to designing operations, interfaces, and so on to be the same across applications and devices
  - Very rarely the case, based on different designer's preference

The google docs icon



Keypad Numbers Layout

A case of external inconsistency

(a) phones, remote controls

1	2	3
4	5	6
7	8	9
	0	

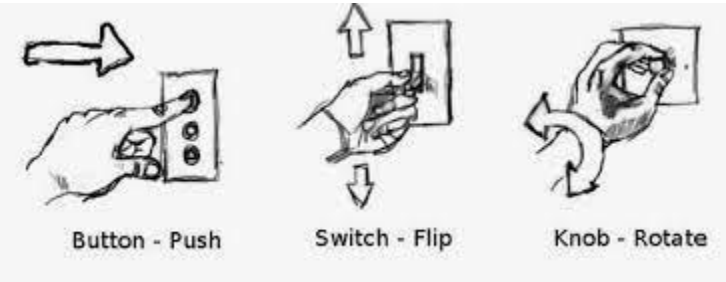
(b) calculators, computer keypads

7	8	9
4	5	6
1	2	3
0		

Affordance: To give a Clue

- Refers to an attribute of an object that allows people to know how to use it. (For example, a mouse button invites pushing, a door handle affords pulling)
- Norman (1988) used the term to discuss the design of everyday objects
- Has since been popularized in interaction design to discuss how to design interface objects (for example, scrollbars to enable moving up and down; icons to click on)
- Affordance also depends on the capability of users. ???

Affordances: To Give a Clue



“Affordances” in Interaction Design?

- Interfaces are virtual and do not have affordances like physical objects
- Norman argues that it does not make sense to talk about interfaces in terms of ‘real’ affordances
- Instead, interfaces are better conceptualized as ‘perceived’ affordances:
  - Learned conventions of arbitrary mappings between action and effect at the interface
  - Some mappings are better than others

One of the main design principles for website design is simplicity. Jakob Nielsen (1999) originally proposed that designers go through all of their design elements and remove them one by one. If a design works just as well without an element, then remove it. Do you think this is a good design principle these days?

Simplicity is certainly an important design principle. Many designers try to cram too much into a screenful of space, making it unwieldy for people to find the information in which they are interested. Removing design elements to see what can be discarded without affecting the overall function of the website can be a salutary lesson. Unnecessary icons, buttons, boxes, lines, graphics, shading, and text can be stripped, leaving a cleaner, crisper, and easier-to-navigate website. However, graphics, shading, coloring, branding and formatting can make a site aesthetically pleasing and enjoyable to use. Good interaction design involves getting the right balance between aesthetic appeal and the optimal amount and kind of information.